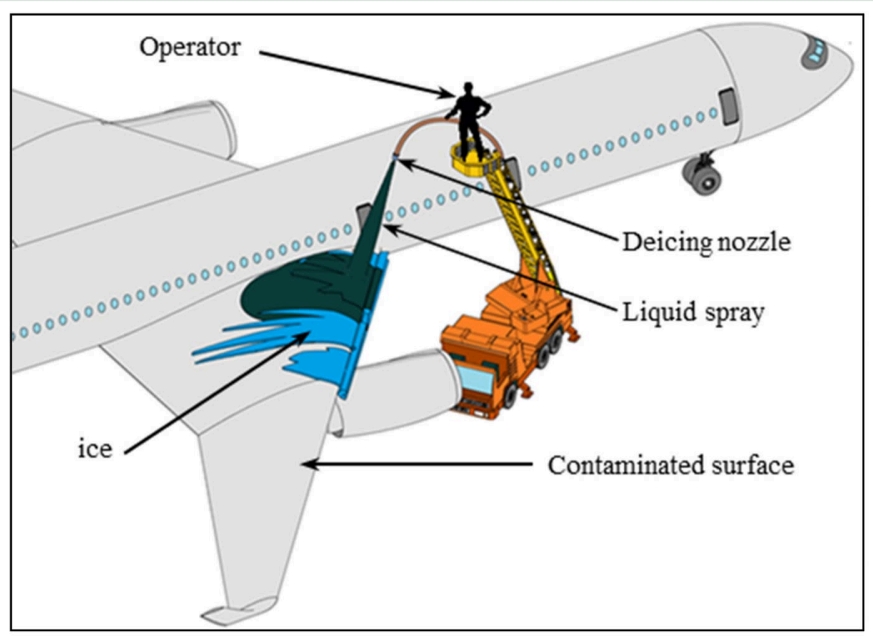
MODULE 2 · DE-ICING AND ANTI-ICING

Lesson 09: Preparing the Aircraft and Protecting Sensitive Areas

Lesson 9 of 30 · about 8 min

LEARNING OBJECTIVES

- Configure the aircraft for de-icing according to the AFM/FCOM.
- Identify areas that must not be sprayed directly.
- Describe good application practice from the crew's point of view.



Application from an elevated basket: the nozzle, the liquid spray and the contaminated surface.

Typical aircraft configuration for treatment

Each aircraft type has a de-icing configuration in the FCOM or the company manuals. Typical items are:

- **Parking brake set** (or brakes held as the pad procedure requires) and the aircraft stationary;
- **Engines** shut down or at idle, as the AFM and the facility procedure require; **APU** running if required;
- **Air conditioning packs and APU bleed off** (or as specified) to prevent fluid fumes entering the cabin and flight deck;
- **Flaps and slats** retracted, unless the AFM requires another setting;
- **Flight controls neutral**, and stabiliser trim as specified;
- **Doors and windows closed**;
- Probe/window heat and other systems as the FCOM specifies.

Confirm the configuration to the de-icing crew before spraying begins (Lesson 10).

Areas that must not be sprayed directly

Area	Why
Pitot probes, static ports, AoA vanes, TAT probes, ice detectors	Fluid residue causes erroneous air data and warnings
Engine inlets, exhausts, thrust reversers; APU inlet and exhaust	Fluid ingestion causes fumes and smoke in the cabin, and possible engine damage
Cockpit and cabin windows	Thickened fluid smears and reduces visibility
Brakes and wheels	Fluid on hot brakes can catch fire or produce smoke
Landing gear bays, electrical connectors and antennas	Contamination of components and connectors
Control surface gaps and balance bays (high-pressure spray)	Forces contamination and fluid into gaps, where residue can collect and freeze

Good application practice

De-icing crews follow their own procedures (SAE AS6285). As a crew member it helps to know what "good" looks like:

- Spraying generally works from the highest point downward and from the leading edge rearward, so that contamination is pushed off the aircraft, not onto clean areas.
- Hot fluid is used close to the surface for de-icing, to transfer heat. Anti-icing fluid is applied as an even, complete layer.
- Both sides of the aircraft are treated in the same way.
- The vertical and horizontal stabilisers, which you cannot see from the flight deck, are treated and checked.
- Treatment of the fuselage depends on the AFM and conditions. On rear-engine aircraft the upper fuselage is a critical surface.

DURING TREATMENT

Do not change configuration, move control surfaces, change engine power or move the aircraft until the de-icing crew confirms that all personnel and equipment are clear.

KEY TAKEAWAYS

- Configure the aircraft exactly as the FCOM specifies before spraying starts.
- Packs and APU bleed off (or as specified) prevent glycol fumes in the cabin.
- No direct spray on probes, engine and APU inlets, windows, brakes or control surface gaps.
- No movement or configuration change until the de-icing crew confirms all clear.

KNOWLEDGE CHECK

Why are the air conditioning packs and APU bleed normally switched off during de-icing?

A. To save fuel

B. To prevent glycol fumes entering the cabin and flight deck

C. To keep the engines warm

D. To reduce noise for the de-icing crew

Answer: B. Fluid drawn into the APU or engine inlets would be carried into the air conditioning system and cause fumes or smoke in the cabin and flight deck.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.